

Toman Bike Share - Data Analytics Portfolio Project

Project Overview

This project involved designing a full-stack data analytics solution for Toman Bike Share, using SQL and Power BI. The goal was to identify revenue trends, profitability drivers, and propose a strategic pricing recommendation for the following year.

Business Requirements

- Hourly Revenue Analysis
- Profit and Revenue Trends
- Seasonal Revenue Performance
- Rider Demographics
- Pricing Strategy Recommendation

Tools Used

- SQL (for data modeling and transformation)
- Power BI (for dashboard creation)
- DAX (for calculated metrics)
- CSV data files representing two years of hourly ride data and cost details

Data Processing and SQL Work

Bike ride data from 2021 and 2022 was combined using a UNION ALL query. This unified dataset was then joined with a cost table to add pricing and cost-per-ride context. Revenue and Profit were calculated using:

$\text{Revenue} = \text{riders} * \text{price}$

$\text{Profit} = \text{riders} * (\text{price} - \text{COGS})$

This logic enabled accurate year-over-year and per-record profitability analysis.

Power BI Dashboard Breakdown

- KPI Cards: Total Riders, Revenue (\$15M), Profit (\$10.45M), and Profit Margin (45%)
- Time-Series Analysis: Riders, Revenue, and Profit trends over 2021-2022

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- Hourly Heatmap: Identifies peak profit hours (10AM-6PM)
- Seasonal Revenue: Summer performed best, suggesting seasonal pricing potential
- Rider Demographics: 81% registered riders supports loyalty-focused strategy

Key Insights & Recommendation

1. Revenue grew 2X between 2021 and 2022 despite a price increase (from \$3.99 to \$4.99)
2. A further 10-15% price increase is viable, suggesting \$5.49 to \$5.74 per ride
3. Peak hour and seasonal demand supports dynamic pricing
4. Casual vs Registered user behavior opens up potential for segmentation

Conclusion

This project demonstrates advanced data transformation with SQL and impactful visual storytelling using Power BI. The dashboard provided actionable insights, helping the business confidently plan pricing strategy for the next year.